pst-bar
Bar charts for pstricks

v.0.93

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With pst-bar, one may use pstricks to produce bar charts directly from a data file. This package is still in the beta stage — the usual caveats pertaining to beta software apply. Additional features and improved (in both content and layout) documentation will be provided as the code stabilizes.

1 Introduction

pst-bar uses the power and flexibility of pstricks [22] to draw bar charts from data stored in a comma-delimited file. Several solutions exist for drawing bar charts for TEX and LATEX documents. The obvious solution is to use an external program, such as gnuplot or Matlab, that can save bar charts in a format that may be read directly into TEX or LATEX. This approach has the disadvantage that extra effort is required to produce charts with a uniform appearance that blend well with the other graphics in a document. In many cases the text in the chart can be typeset by LATEX with the aid of the psfrag or psfragx packages (for diagrams created with Matlab, this process can be accelerated by LaPrint, a Matlab program written specifically for this purpose [10]). However, some programs write the text elements in Encapsulated Postscript files in such a way that the psfrag-based options cannot work.

Bar charts may also be drawn using the built-in drawing tools provided by TEX and LATEX. This is the approach used by the histogr package [12], which is intended to draw histograms, but can be used to draw very simple bar charts. It also appears to be the approach used by the bar package [9] for LATEX 2.09, which defines an environment for drawing bar charts. However, the built-in drawing tools are quite limited and work on the bar package appears to have ceased after 1994.

More sophisticated drawing tools have been developed for use with TEX and LATEX, notably eepic, texdraw and latexdraw, LaTeXPiX, MetaPost, mfpic, and pstricks. Any of these tools might be used to create bar charts; however, none of them has built-in support for doing so, and bar charts must be drawn using the "brute force" method — manually drawing boxes to represent the bars in the chart. Depending on the tool, the axes of the chart might also need to be drawn manually. The pst-bar package is intended to automate the process of drawing bar charts using pstricks.

There are currently two existing solutions for producing bar charts using pstricks. One is the aforementioned brute force method in which a series of \psframes are drawn with suitable heights, widths, and colors to produce the desired chart [16]. While this method works admirably and provides maximum flexibility, it is tedious and labor-intensive. The other option is the bardiag package, which essentially automates this approach [13]. While

effective, bardiag requires a number of external packages to function, uses LaTeX to perform its mathematical operations, and is not compatible with plain TeX.

These issues are all avoided by pst-bar. It calls only on packages that are part of the pstricks family, provides extensive customization features, relies on Postscript for nearly all mathematical operations, and should be compatible with plain TeX (untested). pst-bar provides commands for reading data from a comma-delimited file to produce and label a bar chart. Multiple series of data may be plotted in either clustered or stacked form, and bars may be plotted either horizontally or vertically. An additional option allows block charts that display the differences between subsequent rows of data in the data file (see Section 3.3).

At this stage, pst-bar should be regarded as beta software. Revisions to the code between this release and the first stable release are not guaranteed to be backward compatible.

2 Drawing a Bar Chart

Drawing a bar chart with pst-bar is a three-step process: (i) read the data file, (ii) draw the bars, and (iii) draw the axes. The first two step are accomplished directly with pst-bar; the third requires the pst-plot package, a standard part of pstricks.

The following sections address the data file format, how to load data from the file, and how to use the data to produce a bar chart.

2.1 Data File Format

The data file must be comma-delimited with each row containing the same number of entries. A header row is permitted, but no header entry may contain commas or \par commands (including \\). One of the data files used for the examples in this manual looks like

```
Set 1, Set 2, Set 3
1, 2, 3
1, 2, 3
```

In this case, Set 1, Set 2, and Set 3 are the headers. The subsequent rows, hereafter referred to as *data rows*, contain the data associated with these headers. The file is arranged in a pseudo-columnar fashion, so that the data associated with Set 1 are 1 and 1, those with Set 2 are 2 and 2, and so on. After these data are plotted, the bars will be labeled with the text from the header row.

Warning! There are two very important conditions on the data within the file:

- All data outside of the header must be numeric. If there is no header row, all data in the file must be numeric.
- Each row must contain the same number of entries.

Failure to meet these conditions may result in Postscript errors or (less frequently) incorrectly labeled or drawn bar charts.

How these data are plotted depends on the type of plot desired. See Sections 3.1–3.3 and the examples in Section 4 for more on this.

2.2 Reading the Data File

Use the command \readpsbardata to read from a data file:

```
\readpsbardata[<options>]{<data>}{<filename>}
```

For the data, simply specify a macro name not currently in use.

The available options are:

Parameter	Definition
header	true if first line of data file is header row false otherwise Default: true

If header is set to false, pst-bar will assume that no header is present and treat the first line of the file as data.

2.3 Drawing the Bars

The command \psbarchart draws the bars for the bar chart.

```
\psbarchart[<options>]{<data>}
```

For <data>, use the macro name supplied to \readpsbardata. Bars are grouped in columns that are one \psxunit wide, and the number of bars appearing in a single column is equal to the number of data rows in the data file. It labels the bottom of each column with the headers from the data file if header was set to to true during \readpsbardata. It only creates the bars and the column labels — axes and frames must be provided separately.

The available options are:

Parameter	Definition
chartstyle	cluster produces a cluster of bars stack stacks the cluster into a single column block prints a "floating" bar with a nonzero minimum Default: cluster
barstyle	Name(s) of bar style(s) to use for each bar Default: {black,darkgray,gray,lightgray,white,red,green,blue}
barcolsep	Factor determining whitespace between columns Default: 0.4
barsep	Factor determining whitespace between bars (cluster and block charts) Default: 0.0
barlabelrot	Angle of rotation of the column labels Default: 0
orientation	vertical for vertically drawn bars horizontal for horizontally drawn bars Default: vertical

The number of barstyles specified must equal or exceed the number of bar segments in each column; otherwise, a Postscript error will result. For the cluster and stack chartstyles, this is equal to the number of data rows in the data file; for the block chartstyle, it is equal to half the number of data rows. Furthermore, when specifying the list of barstyles, the styles must be listed between curly braces, e.g.,

```
\psbarchart [Options] {...}
```

The barstyles available by default are red, green, blue, black, white, gray, lightgray, and darkgray. Each of these plots a bar of the named color with square corners and a black outline. New barstyles may be defined using the \newpsbarstyle command described in Section 2.4.

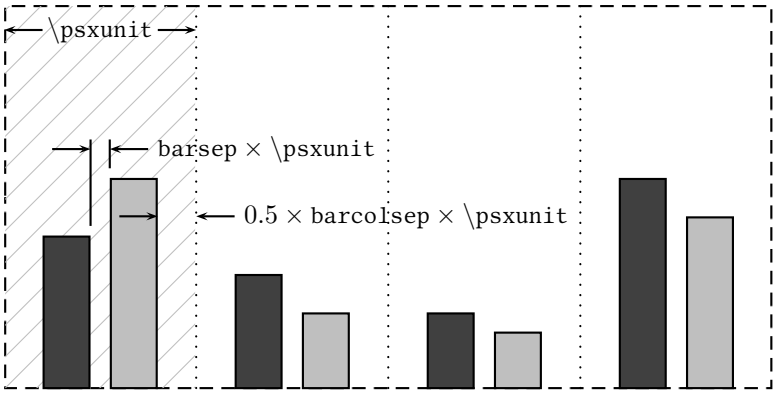


Figure 1: Schematic diagram for the layout of the bar chart. The hatched area marks the region in which data from the first column of the data file will be plotted. Bar width is equal to $[(1 - \texttt{barsep} - 0.5 \times \texttt{barcolsep}) \times \texttt{\psxunit}]/N$, where N is the number of bars in the column.

Figure 1 shows how the `\psbarchart` lays out the bars for the default `chartstyle`, `cluster`. For the `stack` `chartstyle`, the `barsep` option is ignored.

2.4 Customizing the Chart

To add a custom bar style, use the command `\newpsbarstyle`,

```
\newpsbarstyle{<name>}{<definition>}
```

The name is a text string not currently used by any other `barstyle`, and the definition may consist of any `pstricks` key or group of keys applicable to `\psframe`. For example, the `red` `barstyle` is defined as

```
\newpsbarstyle{red}{fillcolor=red ,fillstyle=solid ,framearc=0}
```

To customize the type used to set the column labels, redefine `\psbarlabel`. For example, to have the label set in small italics, use

```
\renewcommand * \psbarlabel{}
```

By default, `\psbarlabel` sets the column label in the current text font.

To adjust the spacing between the bars and the column labels, redefine `\psbarlabelsep`. Note that it is defined as a command, not a length, and should be redefined using `\renewcommand*`. By default it is `0pt`.

Finally, the data from the file may be scaled and manipulated using the command `\psbarscale`,

```
\psbarscale(<scale>){<Postscript code>}
```

inspired by Herbert Voß' `\pstScale` from `pstricks-add` [18]. The data are scaled by the value in parentheses and may be further manipulated with Postscript code. For example, to plot the logarithm of the input data one would use `\psbarscale(1){log}`. The Postscript code is applied to the data prior to scaling.

3 The Chart Styles

As described in Section 2.3, there are three different `chartstyle` options. The default, `cluster` is the common bar chart with bars clustered into groups of related data. The second option,

stack, draws a series of bars stacked atop one another instead of sitting side-by-side in a cluster. The third, block, draws clustered bars between two data points, thus displaying a range of values instead of a single value.

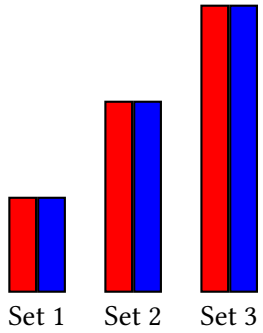
The following sections describe each chartstyle in detail, including how it uses the input data. Examples of use are provided in each case. Full examples, including axes, appear in Section 4.

3.1 cluster

For a cluster chart, each comma-delimited variable within a given row represents a bar in a different column. Each row represents a new set of bars. Thus the file

```
Set 1, Set 2, Set 3
1, 2, 3
1, 2, 3
```

produces the bars



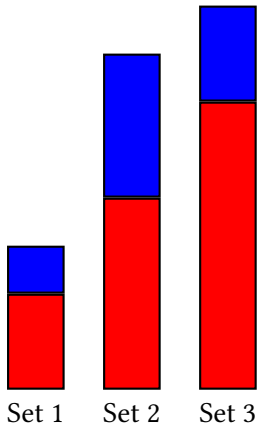

```
\psset{unit=0.5in}%
\begin{pspicture}(0,-0.5)(3,3)%
  \readpsbardata{\data}{example1.csv}%
  \psbarchart[barstyle={red,blue}]{\data}%
\end{pspicture}
```

3.2 stack

For a stack chart, each column of the chart has only one bar. This bar consists of as many segments as the data file has data rows, with the data from each row stacked onto the previous row. Thus, the file

```
Set 1, Set 2, Set 3
1, 2, 3
0.5, 1.5, 1
```

produces the bars




```
\psset{unit=0.5in}%
\begin{pspicture}(0,-0.5)(3,4)%
  \readpsbardata{\data}{example2.csv}%
  \psbarchart[barstyle={red,blue},%
    chartstyle=stack]{\data}%
\end{pspicture}
```

Notice that these bars are wider than those produced by chartstyle=cluster because the bars are stacked vertically instead of being clustered along the bottom axis. A bar width

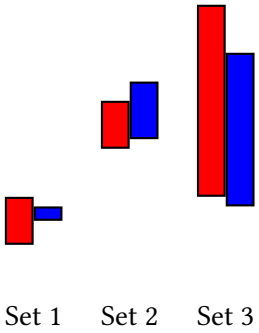
similar to that of the `cluster` chart in the previous section could be obtained by setting `xunit=0.25in`.

3.3 block

For a `block` chart, each bar represents a range of values. As such, each bar requires two data lines from the data file, one denoting the upper limit and the other the lower limit. If there are multiple pairs of data lines, they are plotted in a clustered fashion. If there are an odd number of data lines, the last line of the data file is ignored. Thus, the file

```
Set 1, Set 2, Set 3
1, 2, 3
0.5, 1.5, 1
0.75, 1.6, 0.9
0.9, 2.2, 2.5
```

produces the bars

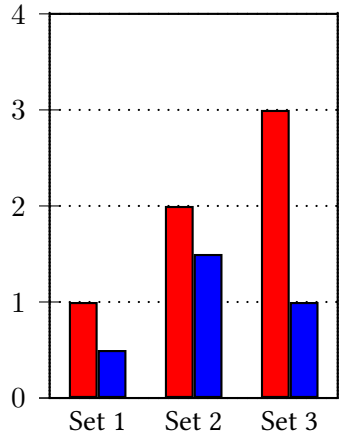



```
\psset{unit=0.5in}%
\begin{pspicture}(0,-0.5)(3,3.5)%
  \readpsbardata{\data}{example3.csv}%
  \psbarchart[barstyle={red,blue},%
    chartstyle=block]{\data}%
\end{pspicture}
```

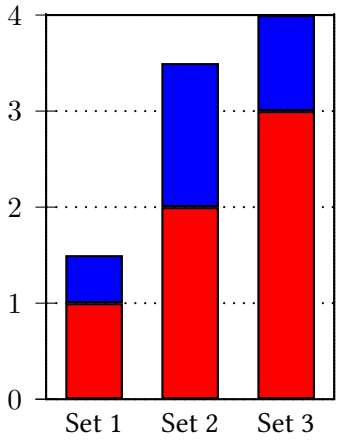
4 Examples

Basic examples of each type including axes and gridlines:

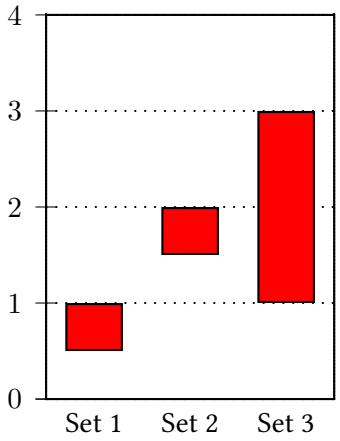
```
\begin{filecontents*}{example2.csv}
  Set 1, Set 2, Set 3
  1, 2, 3
  0.5, 1.5, 1
\end{filecontents*}
```




```
\psset{unit=0.5in}%
\begin{pspicture}(0,-0.5)(3,4.5)%
  \psgrid[xunit=1.5in,gridlabels=0,%
    subgriddiv=0,griddots=30](0,0)(1,4)%
  \psaxes[axesstyle=frame,Ox=0,Dx=1,labels=y,%
    ticks=y](0,0)(3,4)%
  \readpsbardata{\data}{example2.csv}%
  \psbarchart[barstyle={red,blue}]{\data}%
\end{pspicture}
```

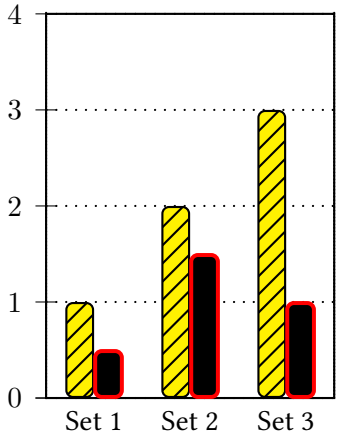



```
\psset{unit=0.5in}%
\begin{pspicture}(0,-0.5)(3,4.5)%
  \psgrid[xunit=1.5in,gridlabels=0,%
    subgriddiv=0,griddots=30](0,0)(1,4)%
  \psaxes[axesstyle=frame,Ox=0,Dx=1,labels=y,%
    ticks=y](0,0)(3,4)%
  \readpsbardata{\data}{example2.csv}%
  \psbarchart[barstyle={red,blue},%
    chartstyle=stack]{\data}%
\end{pspicture}
```



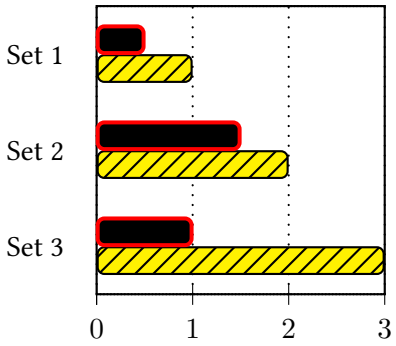

```
\psset{unit=0.5in}%
\begin{pspicture}(0,-0.5)(3,4.5)%
  \psgrid[xunit=1.5in,gridlabels=0,%
    subgriddiv=0,griddots=30](0,0)(1,4)%
  \psaxes[axesstyle=frame,Ox=0,Dx=1,labels=y,%
    ticks=y](0,0)(3,4)%
  \readpsbardata{\data}{example2.csv}%
  \psbarchart[barstyle={red,blue},%
    chartstyle=block]{\data}%
\end{pspicture}
```

Using `\newpsbarstyle`:




```
\psset{unit=0.5in}%
\newpsbarstyle{yellowhatch}{framearc=0.5,%
  fillstyle=hlines*,rot=45,fillcolor=yellow}%
\newpsbarstyle{redoutline}{framearc=0.5,%
  fillcolor=black,linecolor=red,%
  linewidth=1.5pt}%
\begin{pspicture}(0,-0.5)(3,4.5)%
  \psgrid[xunit=1.5in,gridlabels=0,%
    subgriddiv=0,griddots=30](0,0)(1,4)%
  \psaxes[axesstyle=frame,Ox=0,Dx=1,labels=y,%
    ticks=y](0,0)(3,4)%
  \readpsbardata{\data}{example2.csv}%
  \psbarchart[barstyle={yellowhatch,%
    redoutline}]{\data}%
\end{pspicture}
```

Using `[orientation=horizontal]`:




```
\psset{unit=0.5in}%
\newpsbarstyle{yellowhatch}{framearc=0.5,%
  fillstyle=hlines*,rot=45,fillcolor=yellow}%
\newpsbarstyle{redoutline}{framearc=0.5,%
  fillcolor=black,linecolor=red,%
  linewidth=1.5pt}%
\begin{pspicture}(0,-0.5)(3,4.5)%
  \psgrid[xunit=1.5in,gridlabels=0,%
    subgriddiv=0,griddots=30](0,0)(1,4)%
  \psaxes[axesstyle=frame,Ox=0,Dx=1,labels=y,%
    ticks=y](0,0)(3,4)%
  \readpsbardata{\data}{example2.csv}%
  \psbarchart[barstyle={yellowhatch,%
    redoutline}]{\data}%
\end{pspicture}
```

Sophisticated effects are possible using multiple calls to `\psbarchart` in a single chart. For example, a particular important bar may be highlighted in green by splitting the data into two files, one containing a zero for the data value to be highlighted, and the other containing zeros for all of the data *except* the value to be highlighted.

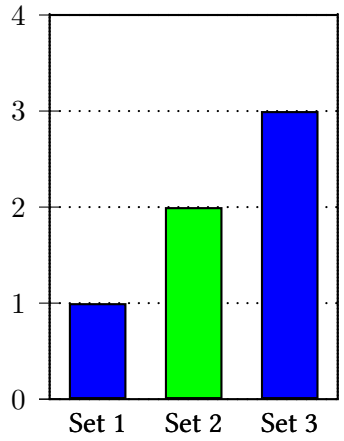
File 1 (`example4.csv`):

```
Set 1, Set 2, Set 3
1, 0, 3
```

File 2 (`example5.csv`):

```
0, 2, 0
```

Notice that the second file has no header row in order to prevent the bar labels from being printed twice.




```
\psset{unit=0.5in}%
\begin{pspicture}(0,-0.5)(3,4.5)%
  \psgrid[xunit=1.5in,gridlabels=0,%
    subgriddiv=0,griddots=30](0,0)(1,4)%
  \psaxes[axesstyle=frame,Ox=0,Dx=1,labels=y,%
    ticks=y](0,0)(3,4)%
  \readpsbardata{\data}{example4.csv}%
  \psbarchart[barstyle={blue}]{\data}%
  \readpsbardata[header=false]{\data}%
    {example5.csv}%
  \psbarchart[barstyle={green}]{\data}%
\end{pspicture}
```

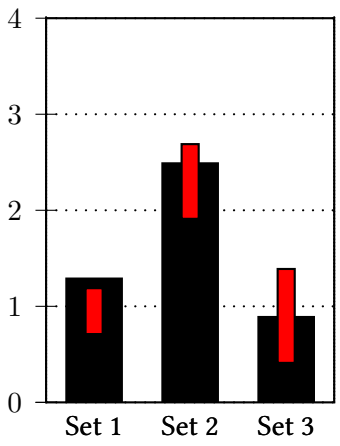
Different chart styles may also be combined in a single bar chart.

File 1 (`example6.csv`):

```
Set 1, Set 2, Set 3
1.3, 2.5, 0.9
```

File 2 (`example7.csv`):

```
0.7, 1.9, 0.4
1.2, 2.7, 1.4
```




```
\psset{unit=0.5in}%
\begin{pspicture}(0,-0.5)(3,4.5)%
  \psgrid[xunit=1.5in,gridlabels=0,%
    subgriddiv=0,griddots=30](0,0)(1,4)%
  \psaxes[axesstyle=frame,Ox=0,Dx=1,labels=y,%
    ticks=y](0,0)(3,4)%
  \readpsbardata{\data}{example6.csv}%
  \psbarchart[barstyle={black}]{\data}%
  \readpsbardata[header=false]{\data}%
    {example7.csv}%
  \psbarchart[barstyle={red},chartstyle=block,%
    barcolsep=0.8]{\data}%
\end{pspicture}
\end{pspicture}
```

5 To Do

- Provide commands to facilitate legend creation and placement.
- Allow the automatic labeling of bars with values from the data file.
- Allow the automatic labeling of bars with labels of the user's choice.
- Improve header parsing to allow commas within header entries.
- Add error-checking to ensure that each row of the data file contains the same number of entries, throwing an error in TEX or LATEX if not.
- Improve documentation.

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